

Project Design Phase

Problem – Solution Fit Template

Date	15 February 2025
Team ID	Individual Project
Project Name	Book - A - Doctor
Maximum Marks	2 Marks

Problem

Patients often face difficulties in finding suitable doctors and booking appointments efficiently. Traditional appointment systems require physical visits or phone calls, leading to long waiting times, scheduling conflicts, and inconvenience. Doctors also struggle to manage appointments manually, resulting in administrative burden and reduced productivity. Additionally, maintaining medical records and appointment history becomes challenging without a centralized digital system.

Target Customers

- Patients seeking medical consultations and healthcare services.
- Doctors who want to manage appointments and patient interactions efficiently.
- Healthcare clinics and small hospitals looking for a streamlined appointment management solution.

Existing Situation

- Appointment booking is often handled manually.

- Patients spend considerable time searching for doctors and available slots.
- Doctors maintain appointment records using traditional methods.
- Communication between patients and doctors is limited.
- Medical reports and appointment history are not easily accessible online.

Customer Pain Points

- Difficulty finding the right doctor.
- Long waiting times for appointments.
- Manual scheduling leads to errors and conflicts.
- Lack of transparency regarding doctor availability.
- Poor management of medical records and appointment history.
- Time-consuming administrative processes for doctors.

Proposed Solution

Book A Doctor is a web-based healthcare appointment management system developed using the MERN Stack. The platform enables patients to register, search for doctors, book appointments, upload medical reports, and track appointment status online. Doctors can manage their profiles, review appointment requests, and approve or reject bookings through a dedicated dashboard.

Key Features

- Patient Registration and Login
- Doctor Registration and Profile Management
- Doctor Search and Filtering
- Online Appointment Booking and Cancellation

- Appointment Approval Workflow
- Medical Report Upload and Management
- JWT-Based Authentication
- Role-Based Access Control
- Appointment Tracking Dashboard

Benefits

For Patients

- Easy access to healthcare services.
- Faster appointment booking process.
- Reduced waiting time.
- Convenient management of appointments and reports.
- Secure access to personal healthcare information.

For Doctors

- Efficient appointment scheduling.
- Reduced administrative workload.
- Better patient management.
- Improved organization of consultation records.

Problem–Solution Fit

The proposed solution directly addresses the challenges faced by both patients and doctors by providing a centralized, secure, and user-friendly platform for healthcare appointment management. The system eliminates manual scheduling issues, improves accessibility, enhances efficiency, and delivers a better healthcare experience for all stakeholders.

Expected Outcome

The Book A Doctor platform will improve healthcare accessibility, simplify appointment management, reduce scheduling conflicts, and provide a modern digital solution that benefits both patients and healthcare providers.